This topic will begin with an introduction to data and information and its relevance in the modern world. You will then learn about decision making, its process and relevance to the application of business analytics.

But before you begin, participate in the discussion below and share your responses with your classmates.

Activity

Decision making

Read the case study 'Entertainment on a cruise ship' by clicking the link. Suppose you were in charge of entertainment management on this ship.

Using the padlet below, answer the questions related to the case study.

Include the following:

1. What does the company want to accomplish and what are the criteria related to entertainment to contribute to business success?
2. How do you describe the problem with entertainment?
3. What are the root causes of the problem according to your evaluation?
4. What data/methods could be helpful to investigate the issues of entertainment and derive insights for improvements?

Once you have posted your answers, read your classmates' answers and 'like' the ones that you agree with and may include points that you may have missed.

To use the padlet

Double click anywhere on the board or tap/click on the pink + to add to the padlet. If you want to open the padlet fullscreen, click on the 'open in new' icon in the top right of the padlet screen. Get creative with photos, videos, LinkedIn profile URLs and more.

Trouble viewing this page? Go to our diagnostics page to see what's wrong.
Skip to content

BINH TRAN + 12M

Share

2025 BUS5PB S1: Week 2: Topic 1 - Decision making

Read the case study 'Entertainment on a cruise ship' by clicking the link. Suppose you were in charge of entertainment management on this ship.

Tracy Nguyen
2 minutes ago

Optimize Entertainment Plan

1) - Things the company wants to accomplish: attract and retain customers with high-quality and diverse entertainment, which will keep passengers happy and engaged throughout the cruise line (especially long cruises 12-15 days).

- Business success criteria:

- Higher attendance and positive feedback from passengers -> Increase overall satisfaction, encourage repeat bookings, and positive word-of-mouth.

- Appealing shows with right-tailored and passenger-focused variety (based on demographics & specific types of shows/performances) -> Ensure different tastes of onboard passengers are met & increase the likelihood of higher attendance.

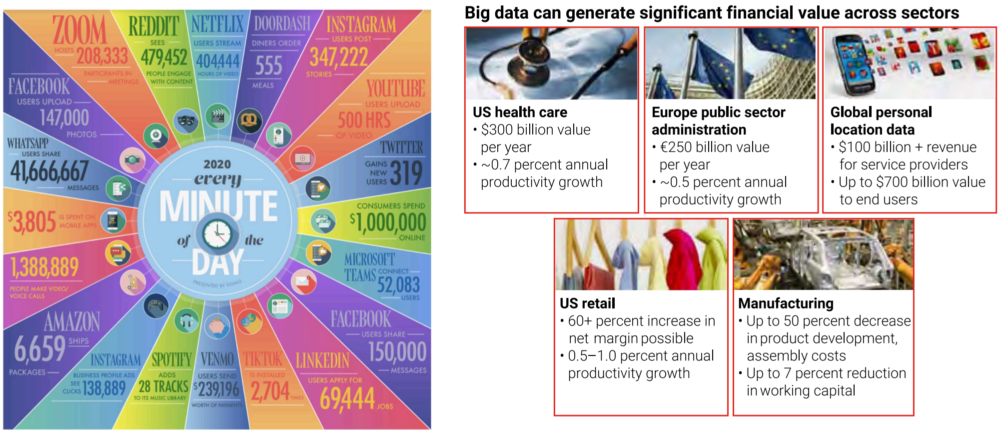
The power of information

Information can be extremely powerful depending on how organisations and individuals utilise it. The world has seen many revolutions, from the agricultural revolution to the industrial revolution and, more recently, the information revolution. We have been experiencing the impact of big data and the challenges posed by big data not just for organisations but also for society at large. With so much data being available, it became challenging to process and investigate this data to understand how it could be used to assist organisations in making informed decisions to bring business value to the organisation.

In Week 1, you looked at some case studies / success stories about how organisations have been using data analytics to improve their business performance. The image below is another example of how much data is being generated and consumed each day, and each one of us is contributing to this data in some form.

Figure 1.1

Daily data generation



(Click on the image to expand)

[Image transcript]

However, it is important to understand how we can use this data to make it a powerful asset for an organisation. This is where data analytics plays an important role as it tells us more about the world that we are living in.

There has been an evolution in the way data and information is being looked at and managed. Data and information have become measurable or quantifiable. This goes back to the late 1940s and early 1950s when the concept of information theory was published. Shannon (1949), who is known as the father of information theory, proved that information can be measured and quantified. In his study, he demonstrated how information could be encoded and decoded for communication, making information an important asset for an organisation.

This has changed the way we view the world and revolutionised how we do business by providing competitive advantages to organisations and making each organisation unique.

To gain such competitive advantage, it is prescribed that every organisation should exploit this uniqueness offered by data analytics and take advantage of the data it has. Every organisation can and should make use of its data (the biggest resource).

Many large organisations are already trying to understand what more than a million of their customers are telling them every day through the internet and social media. It is becoming impossible for companies like Amazon, Google, Meta (Facebook) and eBay to improve business value without understanding what their customers want and without leveraging advanced analytics and sentiment analysis.

Read

This article explains Claude Shannon's information theory. You can read this article to explore and understand more about this theory.

Shannon's information theory (Nguyên Hoang, 2016)

Digital transformation

> *A process that aims to improve an entity by triggering significant changes to its properties through combinations of information, computing, communication, and connectivity technologies.*
>
> *(Gregory, 2019)*

In this definition:

- An entity can be an organisation, society or an industry.
- Transformation is defined as an 'improvement' and it is an 'expected' outcome that is not guaranteed.
- Technology is a combination of information, computing, communication and connectivity.
- A transformation / disruption / fundamental rethink of the use of technology, people and processes to improve or optimise business/organisational performance.
- Examples are paperless offices, on-premise to cloud, digital/social media marketing, lights-out manufacturing.

Digital transformation (DT) is not only about having a website, an app or Twitter account or a chatbot. Digital transformation is applied to answer some of the following questions.

- How does an organisation determine its value proposition in the digital age?
- Who is a digital customer and what is a digital customer experience?
- How does internal organisational culture deliver digital effectiveness?
- What analytics, AI and automation innovations drive digital leadership?

An example of digital transformation in action is shown in a study by Westerman et al. (2014) where they interviewed 157 executives in 50 companies, generating $1 billion or more in annual sales across 15 countries. Half of the interviewees were business leaders such as C suite or managers, and the other half were technology leaders. Technology leaders were invited to answer key questions related to customer experience (CX), operational processes and business models.

Read

This article explains the elements of digital transformation in detail.

The nine elements of digital transformation (Westerman et al., 2014)

Or listen to the audio conversation created by Google NotebookLM to understand these nine elements.

Customer experience

Customer experience includes the following:

Click on the below headings to reveal further information.

Customer understanding

Top-line growth

Top-line growth can be achieved by increase in revenue from core business operations, such as portable devices, sensors, dashboards and personalised portals

Customer touch points

Operational processes

Operational processes include the following:

Click on the below headings to reveal further information.

Process digitisation

Process digitisation includes business process automation, robotic process automation, self-service and chatbot-based engagement

Worker enablement

Performance management

Business models

Business models include the following:

Click on the below headings to reveal further information.

Digital modifications

New digital businesses

Digital globalisation

According to Bonnet and Westerman (2021), digital transformation places emphasis on employee experience (expanded), business model innovation and digital platform (new) as enablers of the other elements and further innovation.

Their model of digital transformation includes the following elements:

- multisided platform businesses
- customer intelligence
- future-readying
- flexforcing.

This is shown in the image below.

Figure 1.2

Digital transformation model

Business model		
Digital enhancements		
Information-based service extensions		
Multisided platform businesses		
Customer experience	**Operations**	**Employee experience**
Experience design	Core process automation	Augmentation
Customer intelligence	Connected and dynamic operations	Future-readying
Emotional engagement	Data-driven decision-making	Flexforcing
Digital platform		
Core		
Externally facing		
Data		

(Click on the image to expand)

[Image transcript]

Read

This article explains the elements of digital transformation in detail.

The new elements of digital transformation (Bonnet & Westerman, 2021)

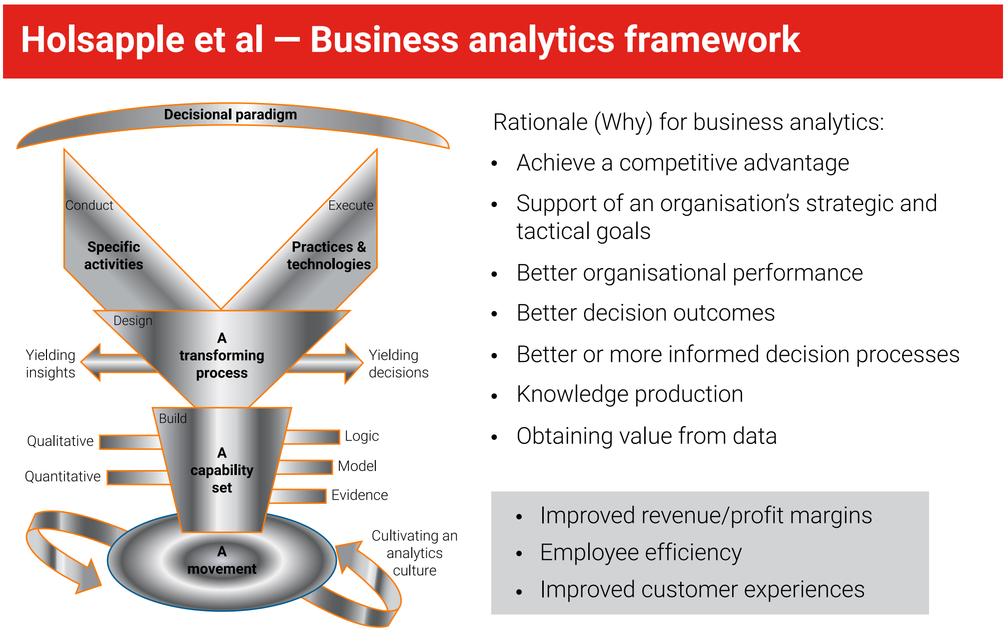
Decision making

To apply analytics, it is important to first understand how analytics can transform or translate to competitive advantage for an organisation. For instance, for an organisation to successfully leverage analytics, it doesn't just need to know the information, it needs to act on the insight it gathers from the information. Execution is essential. This execution will then enable the organisation to drive changes and that is what would bring benefits.

Therefore, decision making is critical to the process of business analytics. A business analytics framework that supports the need to leverage data has been suggested by Holsapple et al. (2014), as shown in the figure below.

Figure 1.3

Business analytics framework



(Click on the image to expand)

© Holsapple et al., 2014, p. 130–141.

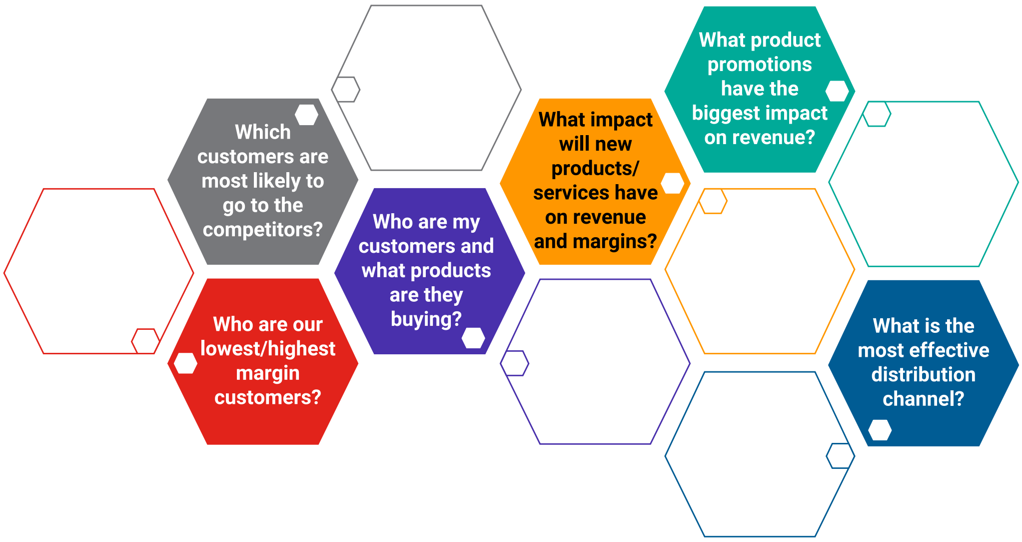
[Image transcript]

This framework illustrates the comprehensive and holistic characterisation of business analytics and provides a rationale for why business analytics is important. Embracing business analytics would allow organisations to improve their revenue and profit margins, improve efficiency and reputation, build goodwill and improve customer experience.

The image below shows some examples of questions that an organisation would need to make decisions on or take actions on based on the insights they uncover from the data.

Figure 1.4

Questions to ask when working with data



(Click on the image to expand)

[Image transcript]

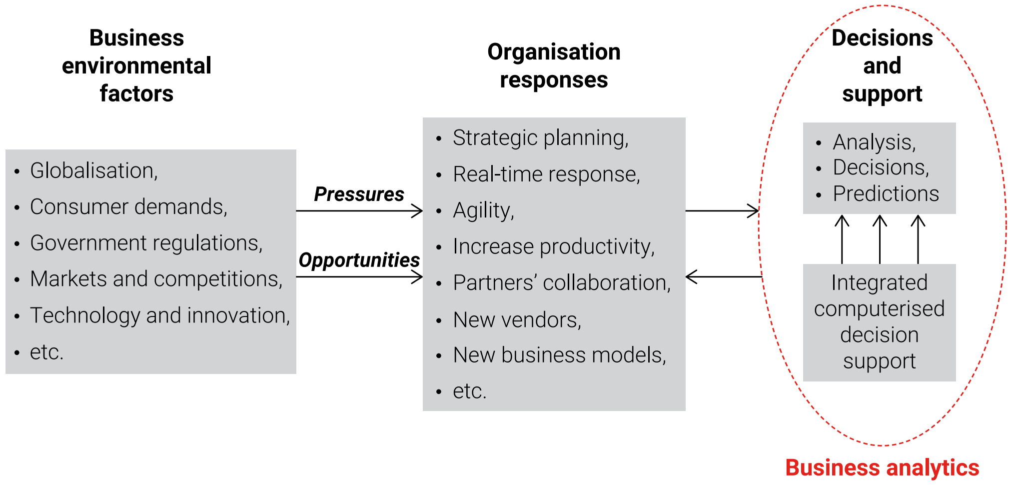
To understand how business analytics can be an enabler to decision making in an organisation and act as a guide in helping organisations make right decisions, we must take into account factors that affect the decision-making process.

As represented in the image below, the various business environmental factors that affect a decision-making process are globalisation, increase or decrease in consumer demand, government regulations, markets and competitions, technology, innovation and so on. These factors lead to pressures that can also be considered as opportunities. The action taken by an organisation on these factors will determine if they are pressures or opportunities. For organisations who look at these factors as opportunities, they would take measures to make the most of it. Some organisational responses to adapt to rapid changes include strategic planning, real-time response, agility, increased productivity, partners' collaboration, new vendors and new business models.

In such a scenario, business analytics supports the organisation's response to these opportunities by helping with the decision-making process through an integrated computerised decision-making system that is supported and validated with data and not based on assumptions. Business analytics facilitates this change by closing the gap between the current performance and the desirable performance of the organisation based on the business goals.

Figure 1.5

The role of business analytics



(Click on the image to expand)

[Image transcript]

Read

This is the full article about the business analytics framework by Holsapple et al (2014).

A unified foundation for business analytics (Holsapple et al., 2014)

Effective decision making

Effective decisions are choices that move an organisation closer to an agreed set of goals in a timely manner.

The three key elements of effective decision making are:

1. a set of goals to work towards
2. means to measure if a chosen course is moving toward or away from those goals
3. information based on those measures must be provided (feedback) to the decision maker in a timely manner.

Multi-level decision making with business intelligence

The assumption can be that only the CEO, president or a chairperson can make decisions. However, this is not valid in the application of business analytics, as a great president may not necessarily understand data analytics.

Absolutely brilliant strategic plans can go wrong because of poor decisions made by those responsible for their implementation. Therefore, it is important to have effective decision makers throughout an organisation. Business analytics requires organisations to move away from traditional structures where junior managers worked only to implement high-level decisions. To ensure success, effective decision making must happen at every level within an organisation.

The three key elements to effective decisions discussed above are applicable to any level of an organisational hierarchy, as shown in the image below. In an ideal organisational set up, the junior-most team members will be responsible for making decisions that affect day-to-day operations, followed by middle management who would look at short-term goals and senior management who would be responsible for making long-term decisions.

Click on the below headings to reveal further information.

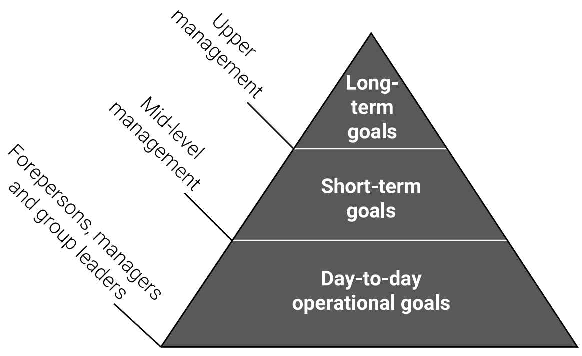
> The top of the pyramid

> Mid-level

> The broad base

Figure 1.6

Figure a: Specific goals at each level of the organisation



[Image transcript]

Figure 1.7

Figure b: Concrete measures at each level of the organisation

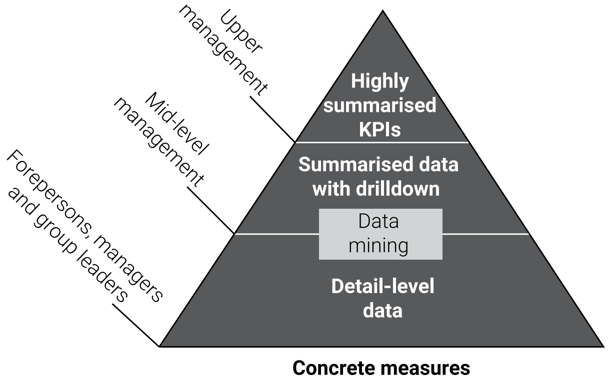


Figure b

[Image transcript]

Figure 1.8

Figure c: Timing of the foundation and feedback information at each level of the organisation

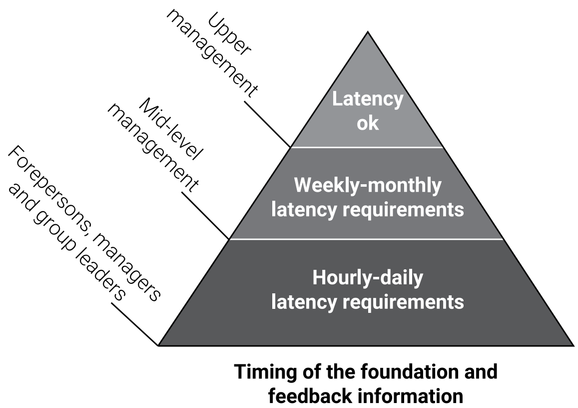


Figure c

[Image transcript]

Some examples of different industries and the kinds of decisions that can be supported by business analytics are:

- banks — credit and risk decisions
- fast-moving consumer goods — supply chain and financial decisions
- fast-food restaurant chain — marketing and performance management decisions
- department store — merchandising and loyalty decisions
- real estate — marketing and finance decisions
- health insurance — claims and disease management decisions
- hospital — critical care decisions and resource management.

Analytics and decision making

There are three different levels of relationship between analytics and decision making. They are:

- automated
- structured human
- loosely coupled.

The level of relationship is the degree of support that is required from analytics to make a decision. The pyramid below explains this degree of support.

In the loosely coupled level, the information obtained from the data platform is not necessary in the decision-making process; that is, it is more likely to be managed by the personnel or the individual within the organisation managing the information.

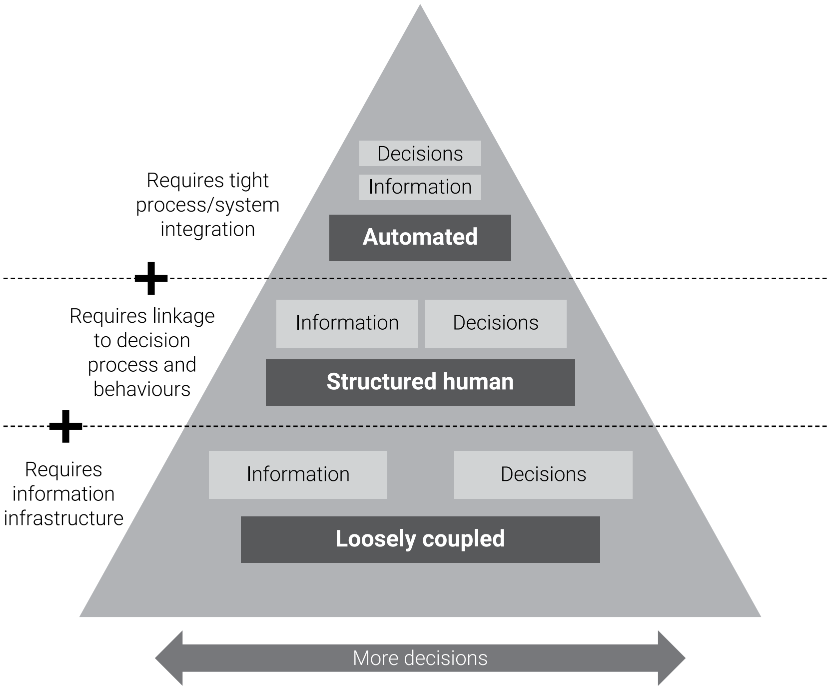
The structured human level is a mix of both loosely coupled and automated, where some decisions are based on data and some managed by the personnel or individual within the organisation.

The automated level or the tightly coupled level is where certain processes or decisions can be automated, but that doesn't mean that no personnel or individual is involved.

You will now explore each of these levels in detail.

Figure 1.9

The different levels of relationship between analytics and decision making



(Click on the image to expand)

[Image transcript]

Click on the below headings to reveal further information.

› Loosely coupled analytics and decisions

› Structured human decision making

› Automated decisions

Activity

Case study

Read the case study 'Holistic marketing approach by big data integration at an insurance company' by clicking the link.

Then, using the padlet below answer the questions related to the case study.

Include the following:

1. What are the challenges the insurance company faced?
2. Referring to the key elements of effective decision making introduced in this topic, explain how the solution facilitated better decision making of the company.

Once you have posted your answers, read your classmates' answers and 'like' the ones that you agree with and may include points that you may have missed.

To use the padlet

Double click anywhere on the board or tap/click on the pink + to add to the padlet. If you want to open the padlet fullscreen, click on the 'open in new' icon in the top right of the padlet screen. Get creative with photos, videos, LinkedIn profile URLs and more.

Trouble viewing this page? Go to our diagnostics page to see what's wrong.
Skip to content

BINH TRAN 3MO

Share

2025 BUS5PB S1: Week 2: Topic 1 - Case study

Read the case study 'Holistic marketing approach by big data integration at an insurance company' by clicking the link.

Made with Padlet

Last modified: Wednesday, 5 March 2025, 3:31 PM

← References

Jump to...

Topic 2: Analytics and strategy →